



**UNIMORE**

UNIVERSITÀ DEGLI STUDI DI  
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria  
"Enzo Ferrari"

# Processi di deformazione plastica

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# Introduzione

Tecniche di **lavorazione primaria** che permettono di ottenere manufatti di geometria relativamente semplice mediante deformazione plastica del metallo.



# Introduzione

Il materiale di partenza può essere in forma di piastre, lamiera, barre, filo o tubi a diversa sezione.

Esistono 3 classi principali di processo:

- a) **FORGIATURA**: produzione di un componente discreto (numerabile) mediante la compressione del metallo in una serie di stampi. Si esegue di solito a caldo.

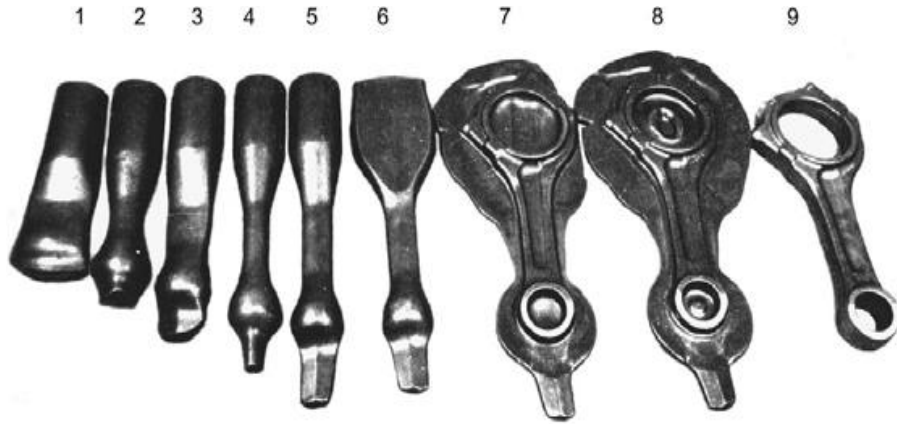
Approfondimento: <https://www.dropforging.net/>

- b) **ESTRUSIONE**: produzione di componenti continui (a metri) a sezione costante, solidi o forati. Il prodotto continuo è poi tagliato alla lunghezza desiderata. Eseguita sia a caldo che a freddo.

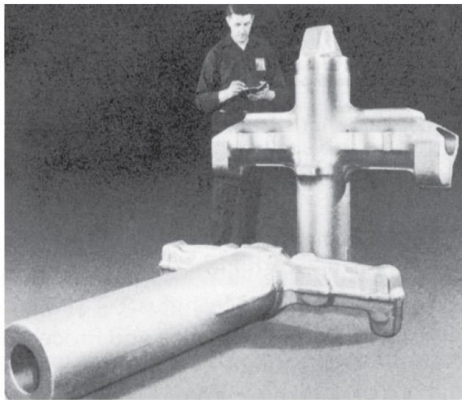
Approfondimento: <https://www.sdql-alu.com/>

- c) **Processi di deformazione delle LAMIERE**: classe di processi riservati alla sola trasformazione delle lamiere.

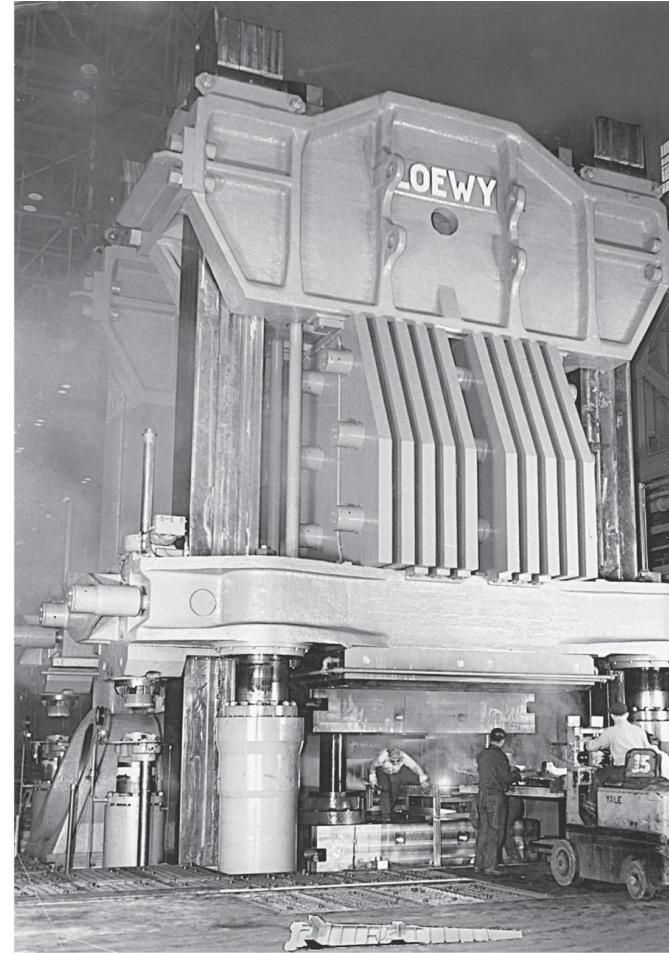
# Forgiatura



(a)



(b)



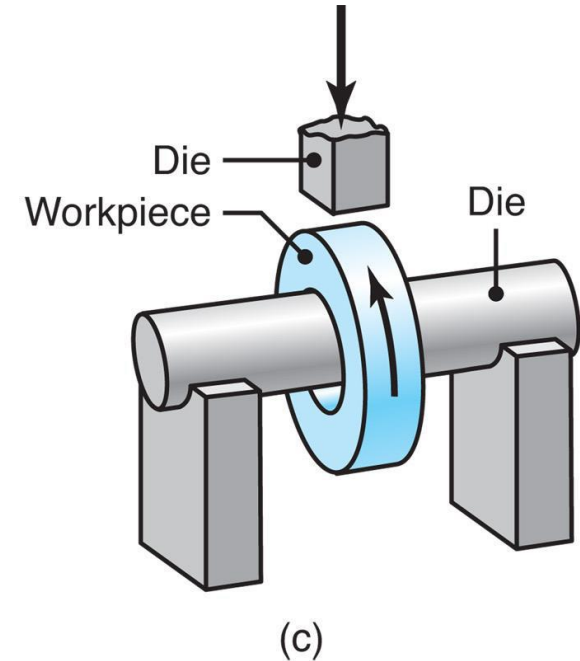
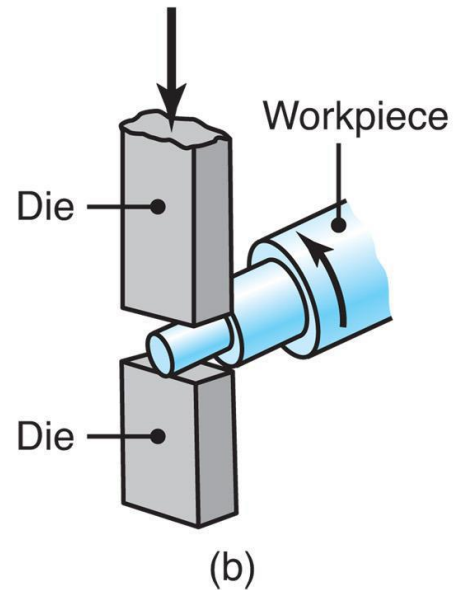
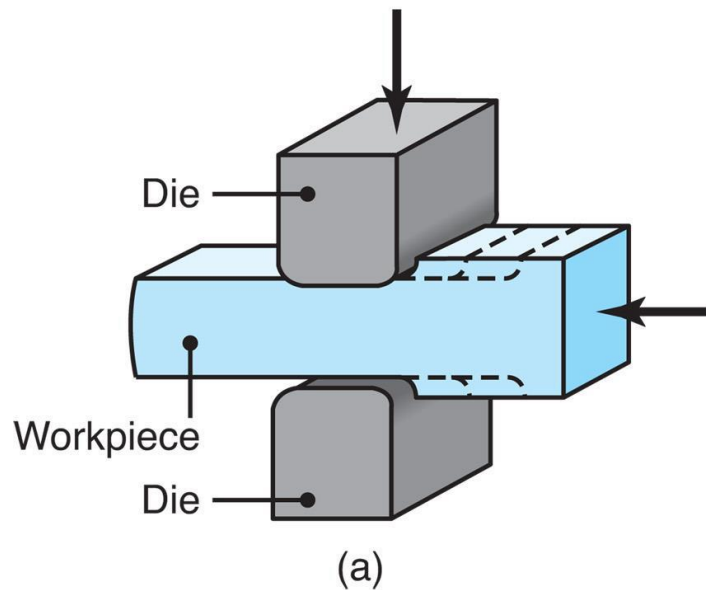
(c)

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[Video: stampo aperto](#)

# Forgiatura

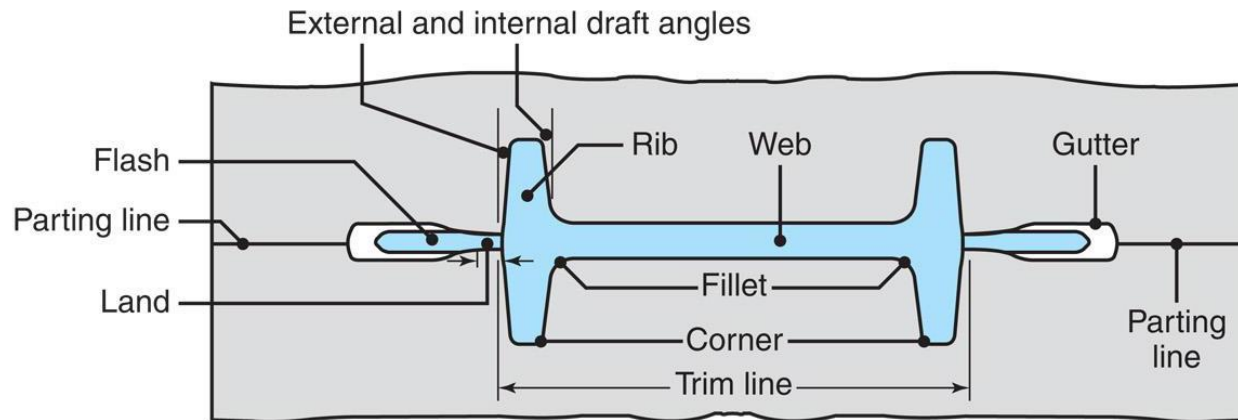
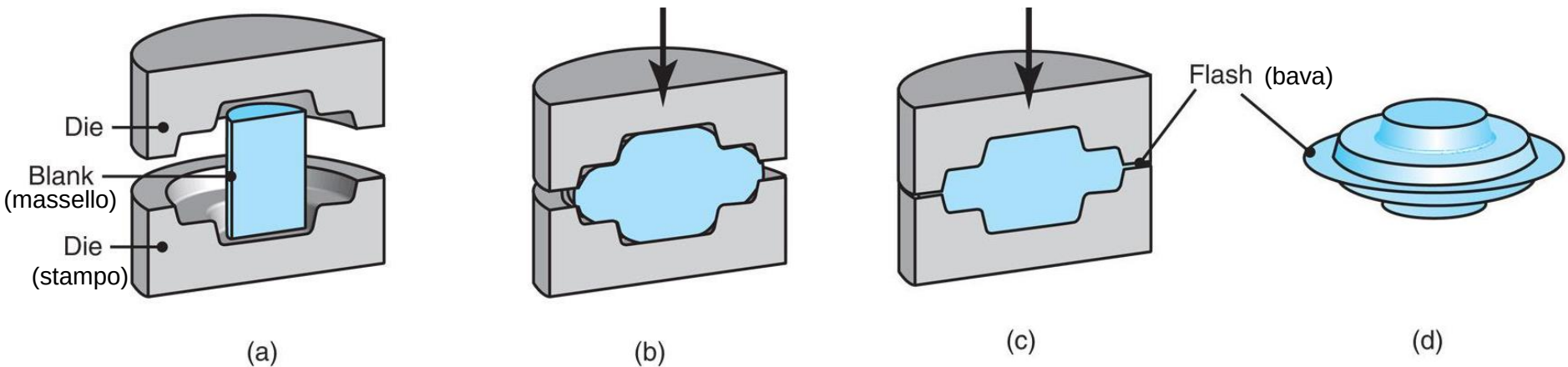
## Esempi forgiatura in stampo aperto



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# Forgiatura

## Schema forgiatura in stampo chiuso



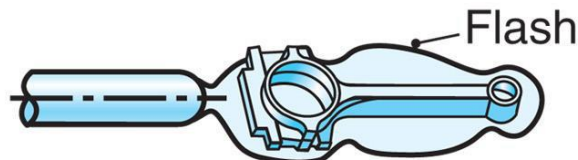
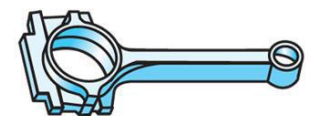
# Forgiatura



[Video: stampo chiuso](#)

[Video: gear forging](#)

# Forgiatura

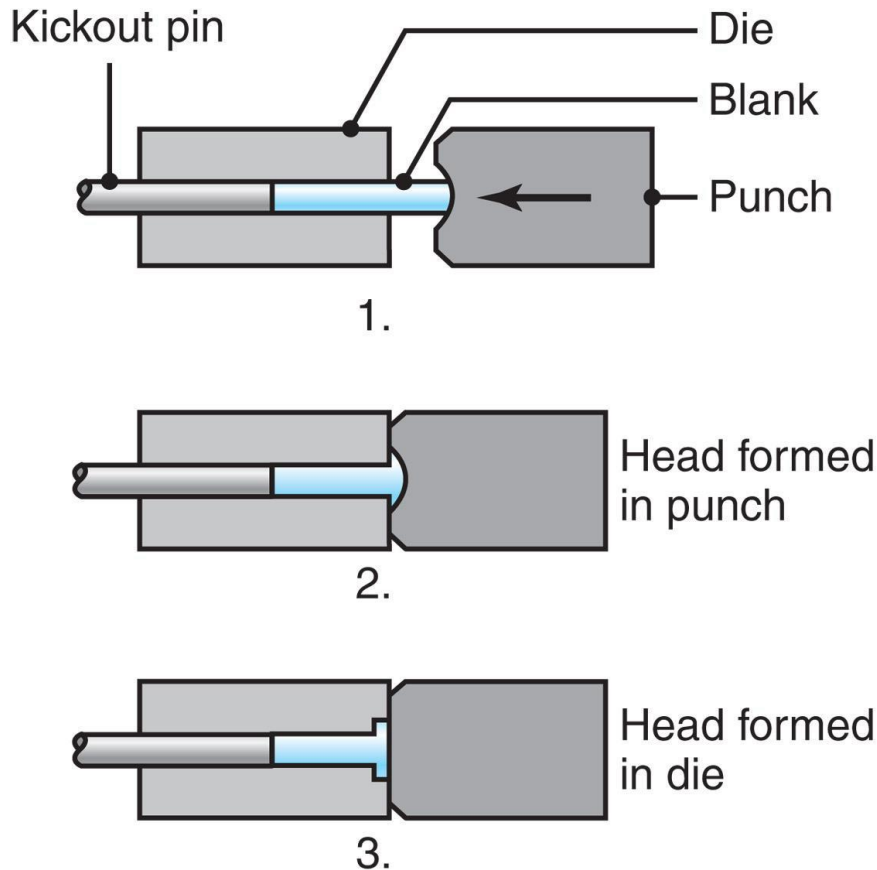
1.   
Blank (bar stock)
2.   
Edging
3.   
Blocking
4.   
Finishing
5.   
Trimming

# Forgiatura

## General Characteristics of Forging Processes

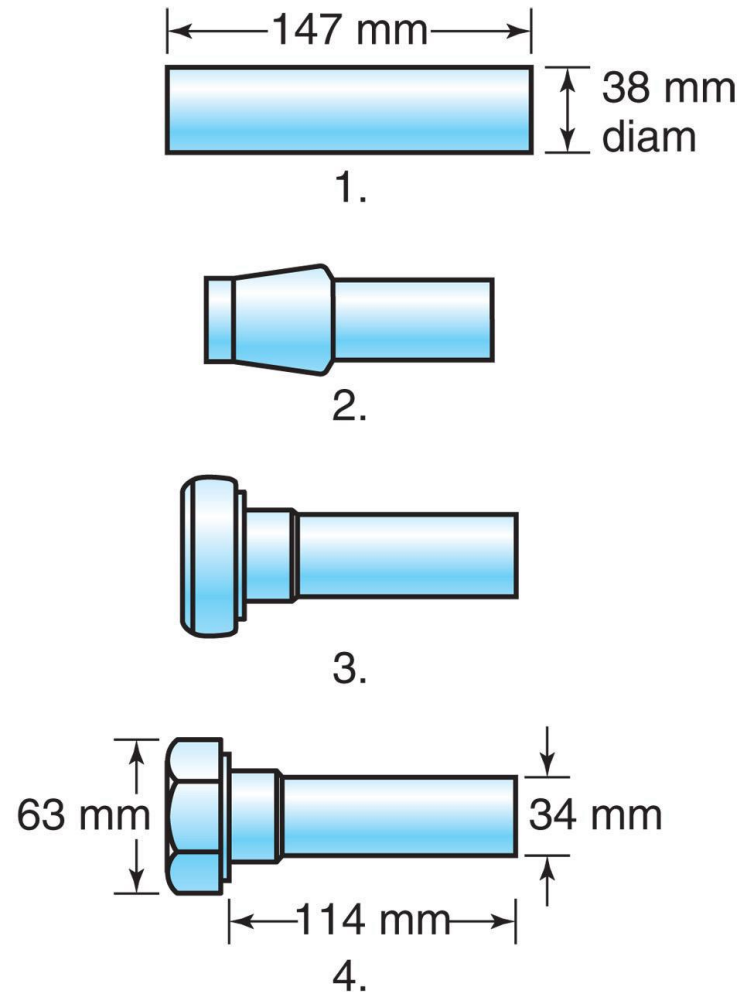
| Process    | Advantages                                                                                                                                                          | Limitations                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open die   | Simple and inexpensive dies; wide range of part sizes; good strength characteristics; generally for small quantities                                                | Limited to simple shapes; difficult to hold close tolerances; machining to final shape necessary; low production rate; relatively poor utilization of material; high degree of skill required |
| Closed die | Relatively good utilization of material; generally better properties than open-die forgings; good dimensional accuracy; high production rates; good reproducibility | High die cost, not economical for small quantities; machining often necessary                                                                                                                 |
| Precision  | Close dimensional tolerances; very thin webs and flanges possible; machining generally not necessary; very good material utilization                                | High forging forces, intricate dies, and provision for removing forging from dies                                                                                                             |

# Forgiatura



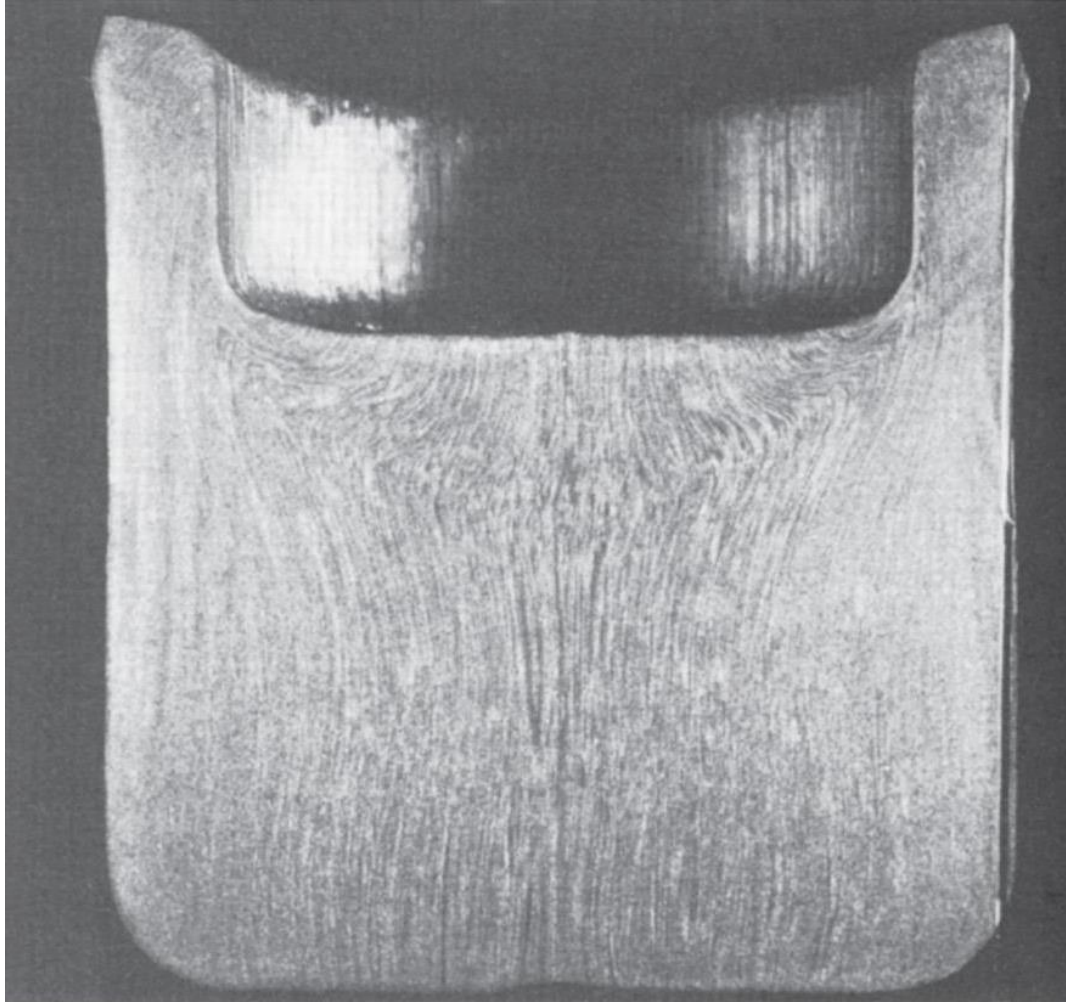
Modalità per la realizzazione  
della testa di una vite

[Video](#)

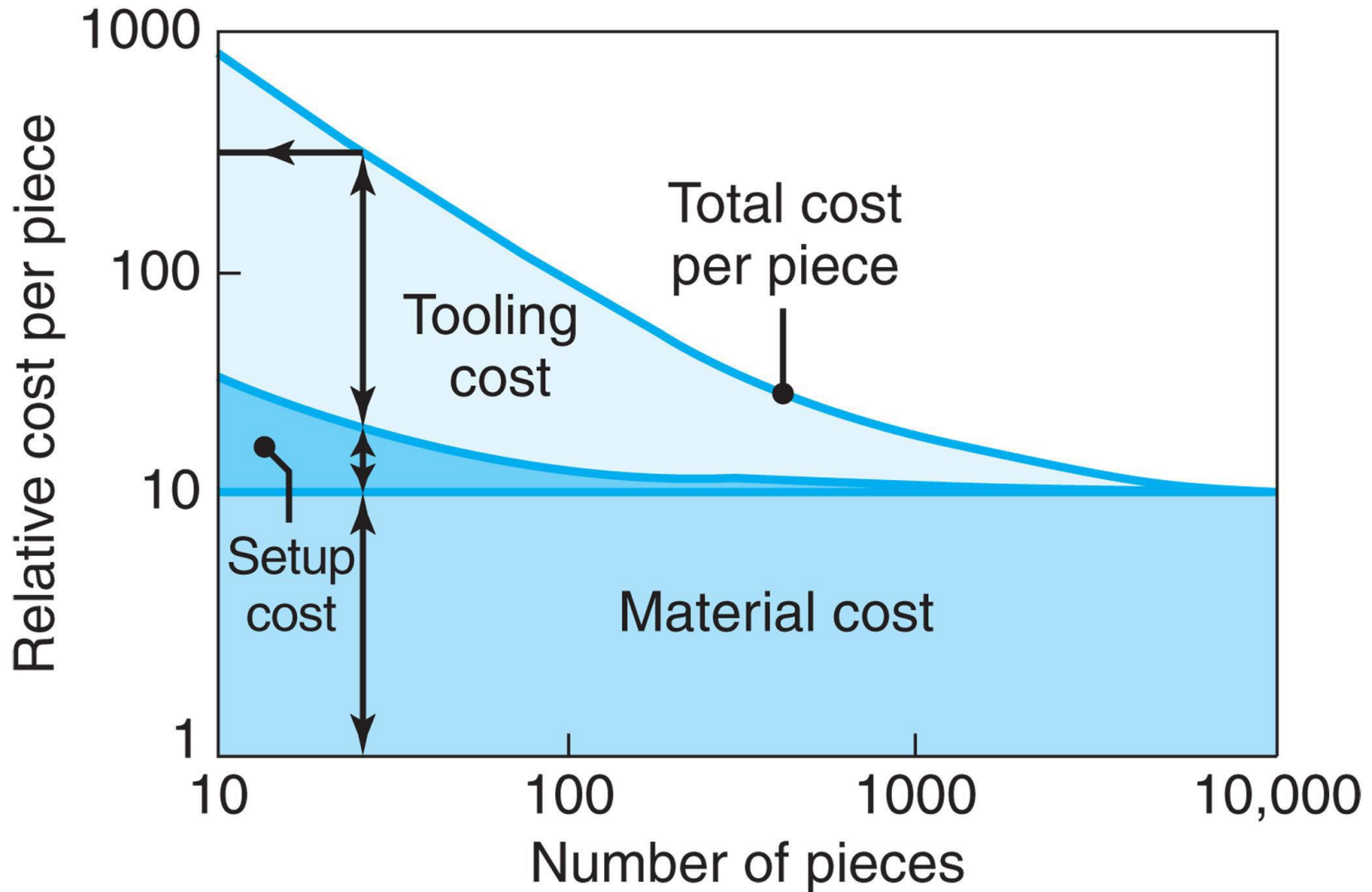


Sequenza di operazioni per  
l'ottenimento della testa di una vite

# Forgiatura



# Forgiatura



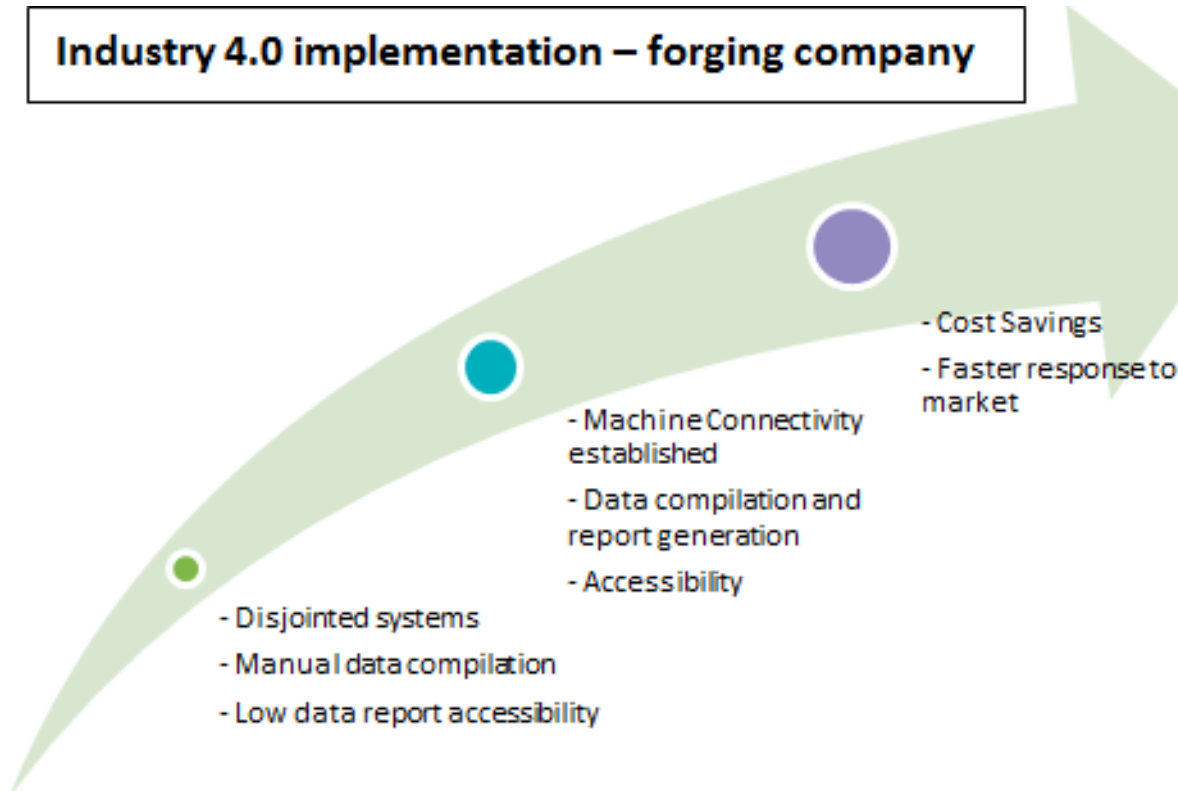
The graph illustrates the relative cost per piece for five manufacturing processes as a function of the number of pieces produced. The y-axis represents the relative cost per piece on a logarithmic scale from 0.1 to 100. The x-axis represents the number of pieces on a logarithmic scale from 100 to 100,000. The processes and their approximate cost curves are:

- Sand casting:** Starts at a relative cost of approximately 2.0 for 100 pieces and decreases to about 0.15 for 100,000 pieces.
- Gravity die-casting:** Starts at a relative cost of approximately 1.5 for 100 pieces and decreases to about 0.3 for 100,000 pieces.
- Forging:** Starts at a relative cost of approximately 25 for 100 pieces and decreases to about 0.3 for 100,000 pieces.
- Investment casting:** Starts at a relative cost of approximately 8 for 100 pieces and decreases to about 0.3 for 100,000 pieces.
- Pressure die-casting:** Starts at a relative cost of approximately 8 for 100 pieces and decreases to about 0.3 for 100,000 pieces.

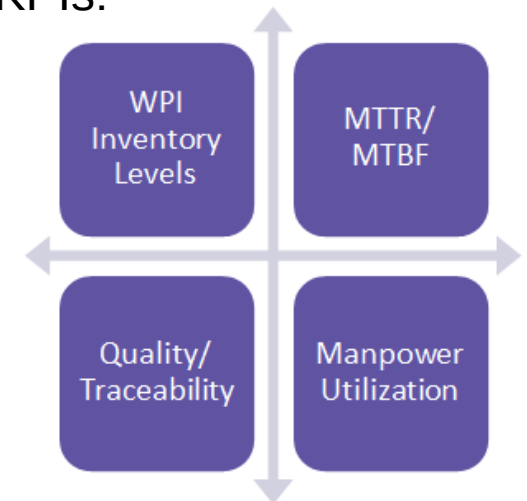
Two diagrams of a mechanical part are shown at the top. The left diagram is a cross-section with a width of 16 mm. The right diagram is a side view with a length of 73 mm.

# Industria 4.0 applicata alla forgiatura

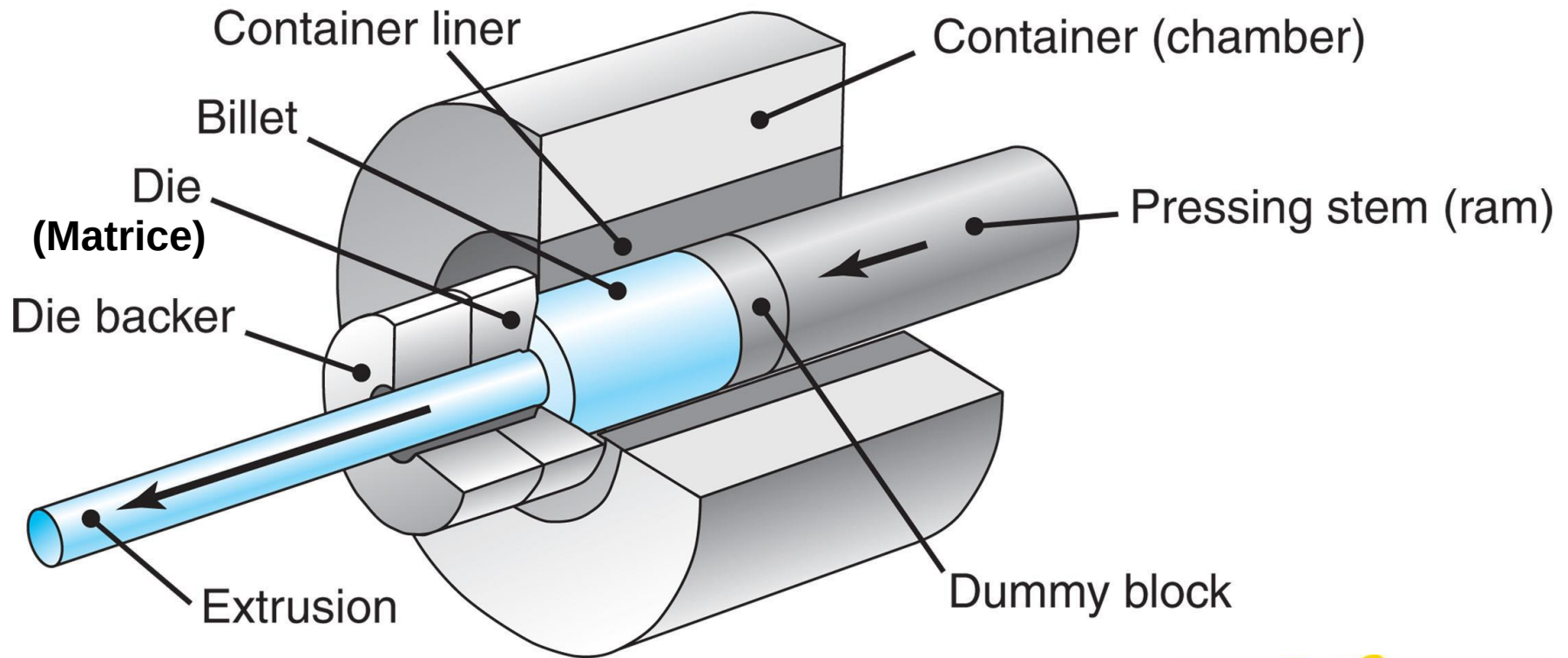
## Industry 4.0 implementation – forging company



### KPIs:



# Estrusione



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[Video 1](#)

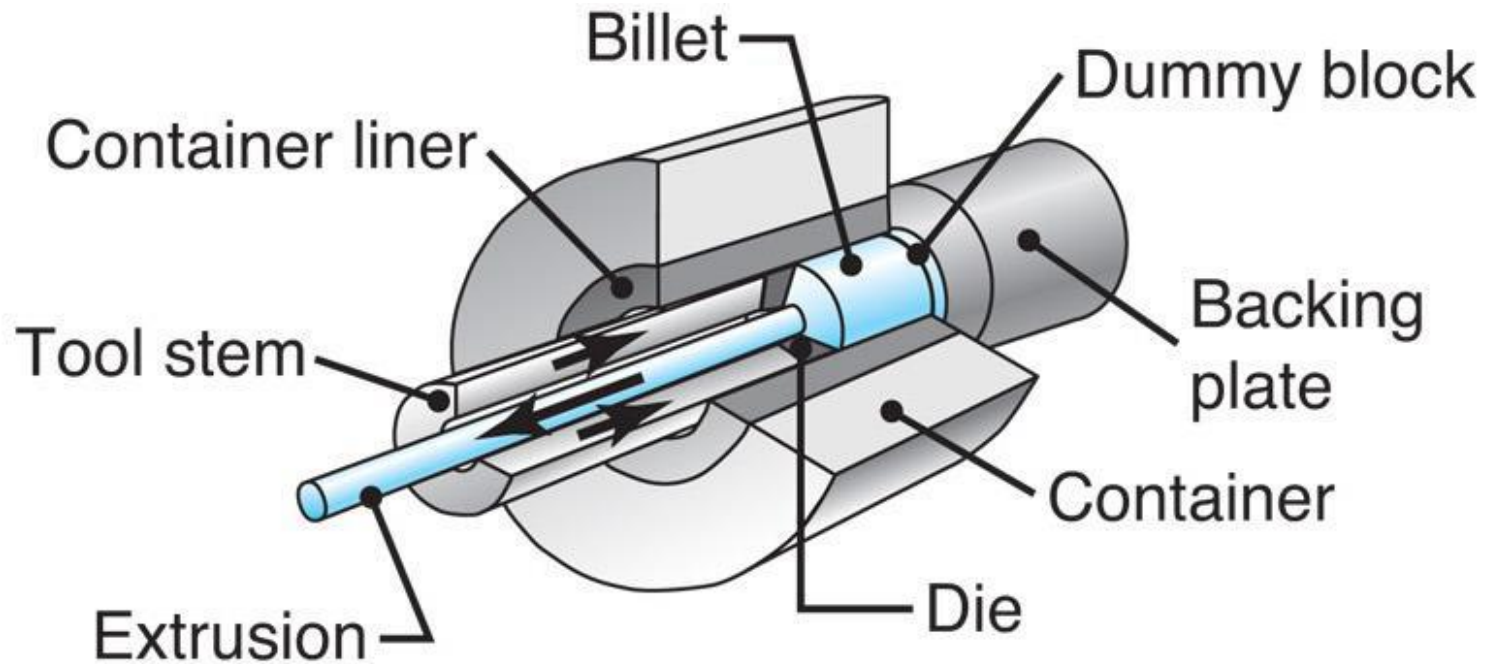
[Video 2](#)

[Animazione impianto completo](#)



# Estrusione

## Estrusione indiretta o inversa



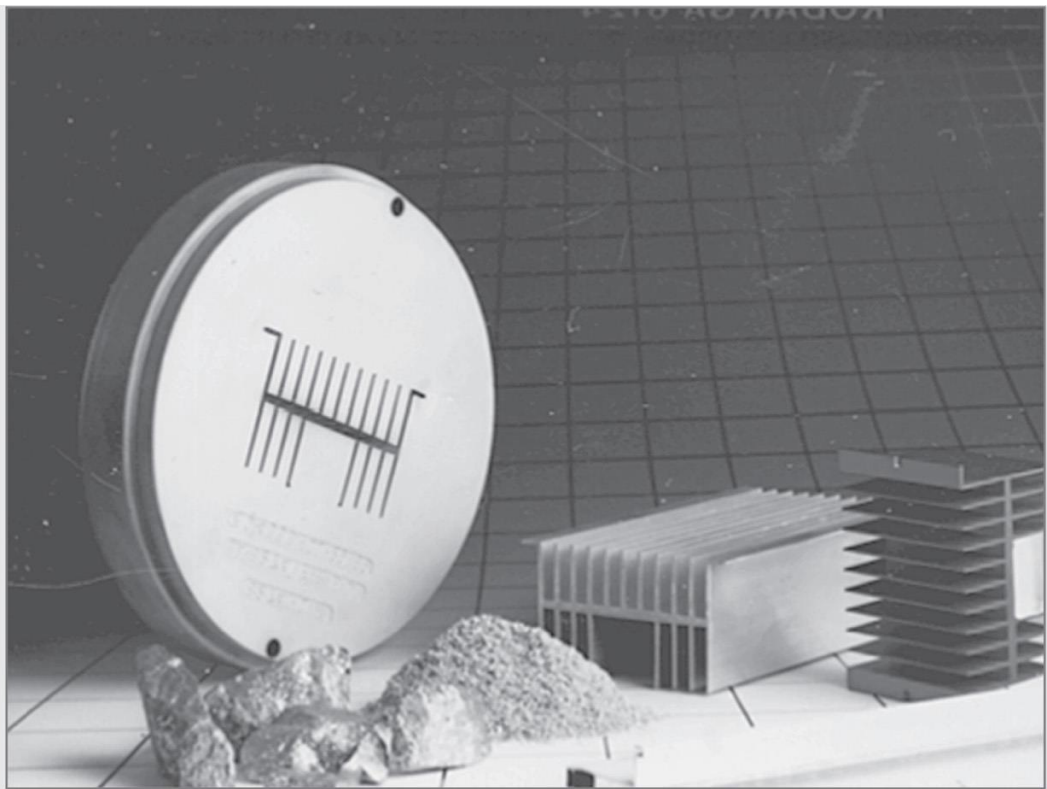
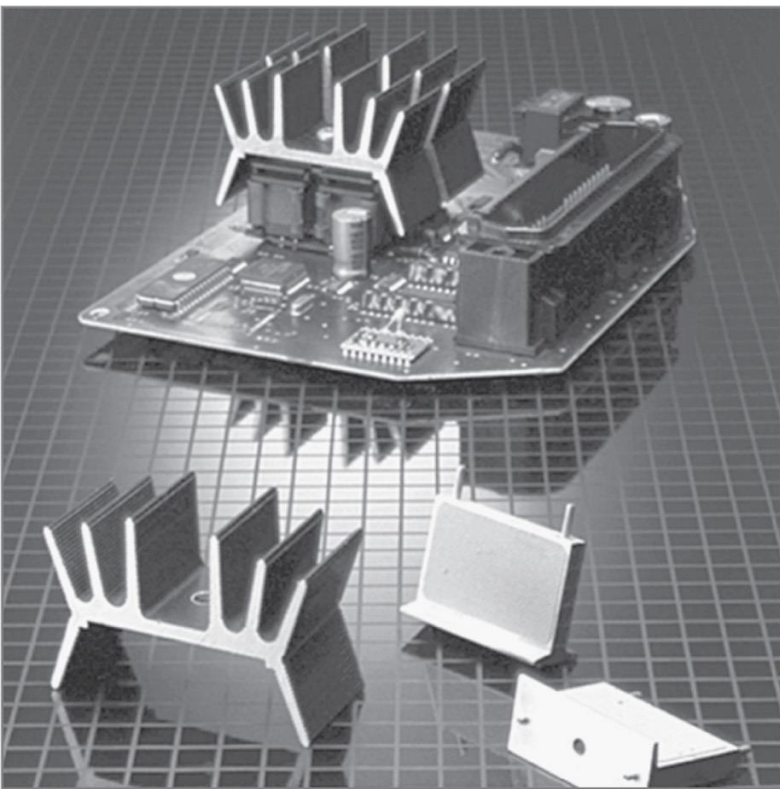
# Estrusione

Applicazioni:

- Serramenti
- Telai per macchinari
- Dissipatori di calore
- Elementi strutturali



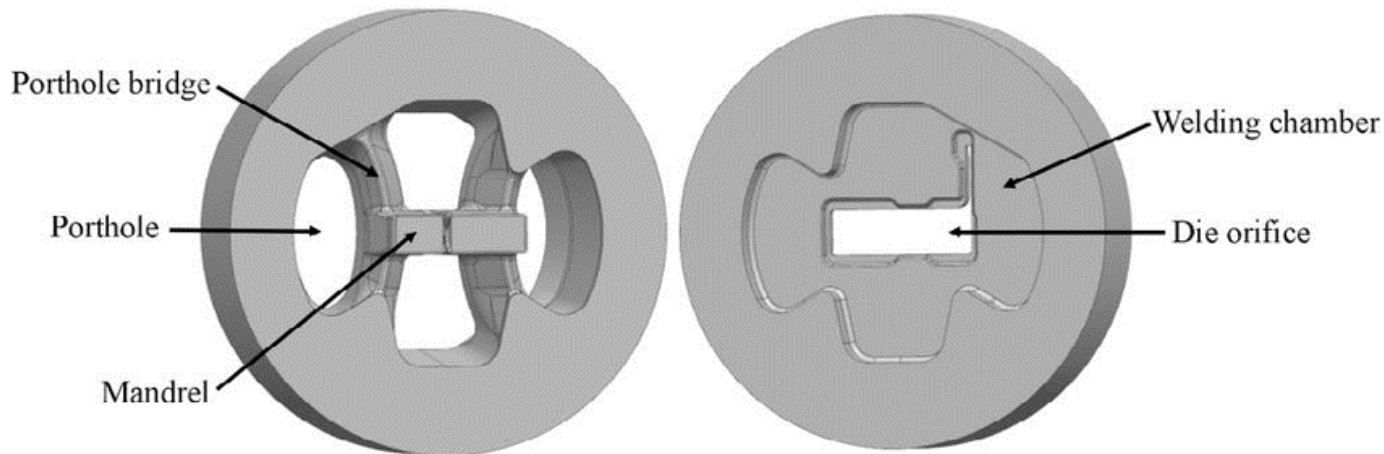
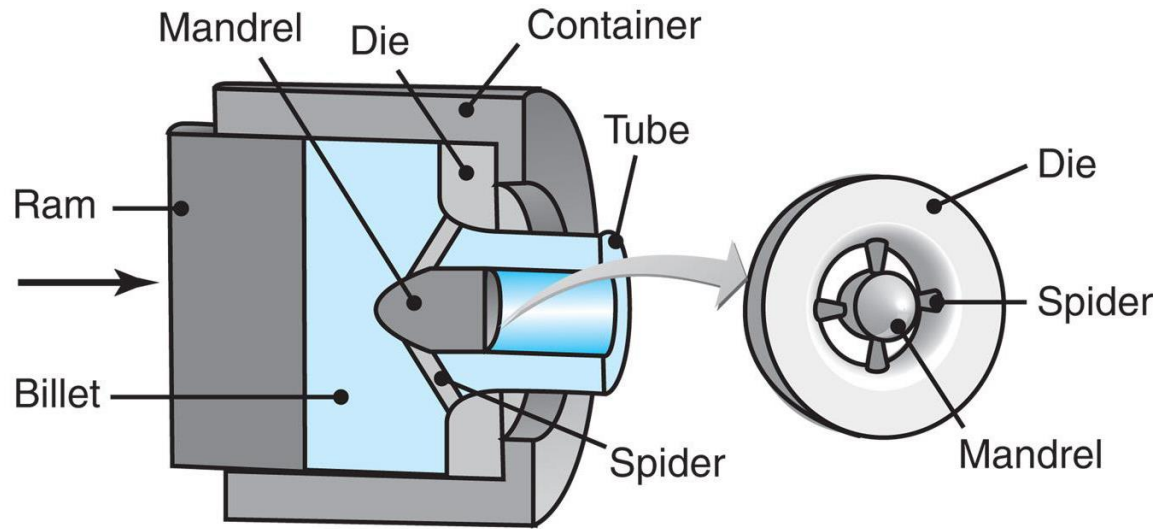
# Estrusione



[Video](#)

# Estrusione

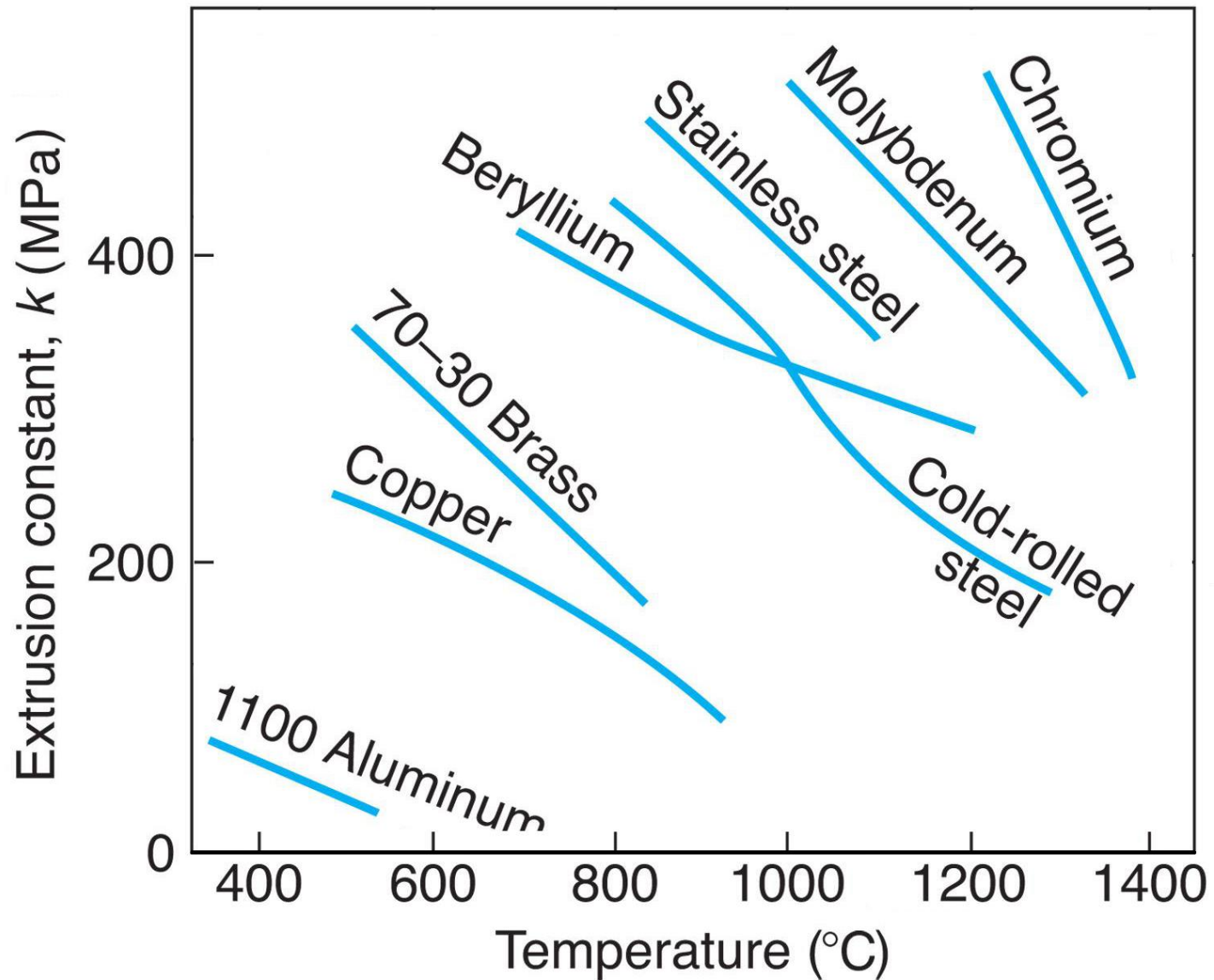
## Ottenimento di prodotti cavi



# Estrusione



# Estrusione



# Industria 4.0 applicata all'estrusione

[Source](#)



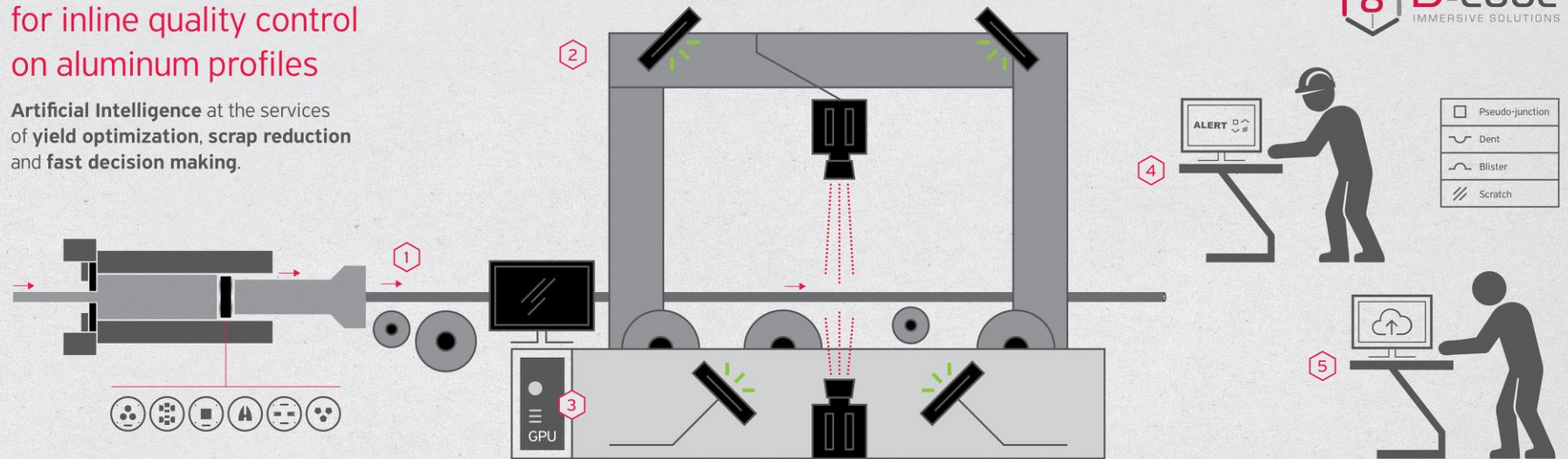
[Video](#)



# Industria 4.0 applicata all'estrusione

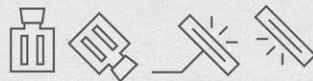
## Machine Vision System for inline quality control on aluminum profiles

**Artificial Intelligence** at the services  
of **yield optimization, scrap reduction**  
and **fast decision making**.



1. Profile exits extrusion press on the run-out roller conveyor
2. Profile enters CYRUS Smart Cage, just a few meters from the press
3. CYRUS captures extruded profile and artificial intelligence algorithms identify surficial defects in less than 50msec
4. Press Operator receives real-time scrap alerts and is instructed to tackle press parameters
5. Production manager monitors remotely and near real-time the production and has access to big data analytics, enabling yield optimization.

Cyber Physical Aluminum Extrusion (CYRUS) System using high-end GPUs and neural networks for surface inspection and early defect detection, in less than 50msec.



Basler industrial cameras with seamless recording style and powerful industrial LED lighting systems from Efflux, allow for early defect detection and localization in speeding profiles.



Alert press operator to frequent and repetitive blisters, dents, welds, scratches, oxides, skews and/or emerging die lines, in real time.

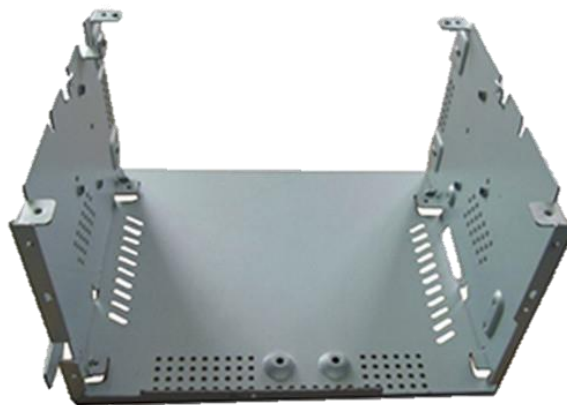


Big data on the cloud and Infrastructure-as-a-Service, reveal insights for yield optimization and bring aluminum business to the next level.

[Source](#)

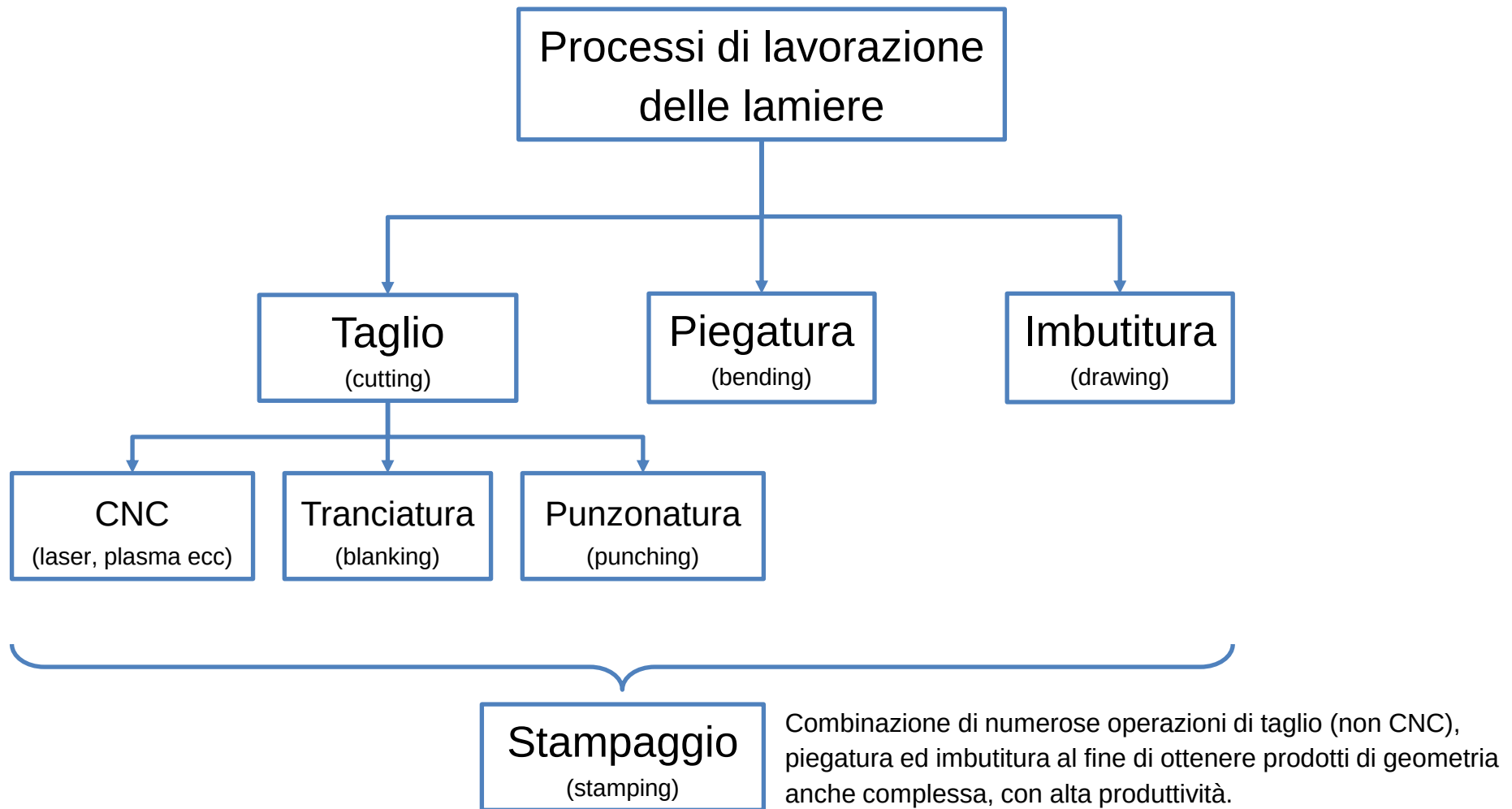
# Lavorazione delle lamiere

**Lamiera:** semilavorato caratterizzato da un'elevata estensione in due dimensioni e da uno spessore sottile (0.1 – 5 mm circa). Sono fornite alle aziende in forma di bobine.



[Video](#)

# Lavorazione delle lamiere



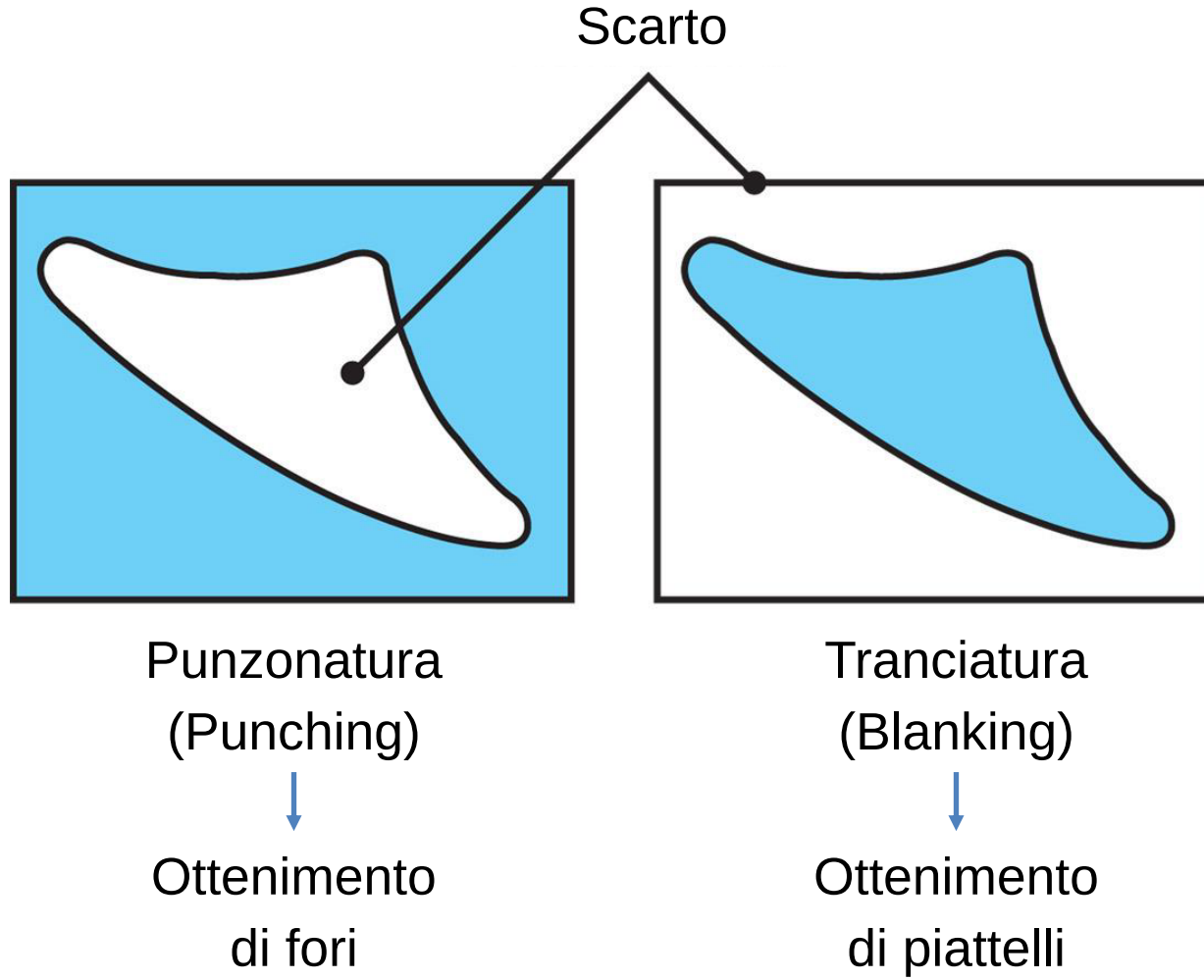
# Lavorazione delle lamiere

## Esempi di lavorazioni applicate alle lamiere per conferire una geometria 3D

| Forming process      | Characteristics                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drawing (Imbutitura) | Shallow or deep parts with relatively simple shapes, high production rates, high tooling and equipment costs                                                                                                                           |
| Explosive            | Large sheets with relatively simple shapes, low tooling costs but high labor cost, low-quantity production, long cycle times                                                                                                           |
| Incremental          | Simple to moderately complex shapes with good surface finish; low production rates, but no dedicated tooling required; limited materials                                                                                               |
| Magnetic-pulse       | Shallow forming, bulging, and embossing operations on relatively low strength sheets, requires special tooling                                                                                                                         |
| Peen                 | Shallow contours on large sheets, flexibility of operation, generally high equipment costs, process also used for straightening formed parts                                                                                           |
| Roll                 | Long parts with constant simple or complex cross-sections, good surface finish, high production rates, high tooling costs                                                                                                              |
| Rubber               | Drawing and embossing of simple or relatively complex shapes, sheet surface protected by rubber membranes, flexibility of operation, low tooling costs                                                                                 |
| Spinning             | Small or large axisymmetric parts; good surface finish; low tooling costs, but labor costs can be high unless operations are automated                                                                                                 |
| Stamping             | Includes a wide variety of operations, such as punching, blanking, embossing, bending, flanging, and coining; simple or complex shapes formed at high production rates; tooling and equipment costs can be high, but labor cost is low |
| Stretch              | Large parts with shallow contours, low-quantity production, high labor costs, tooling and equipment costs increase with part size                                                                                                      |
| Superplastic         | Complex shapes, fine detail and close dimensional tolerances, long forming times (hence production rates are low), parts not suitable for high-temperature use                                                                         |

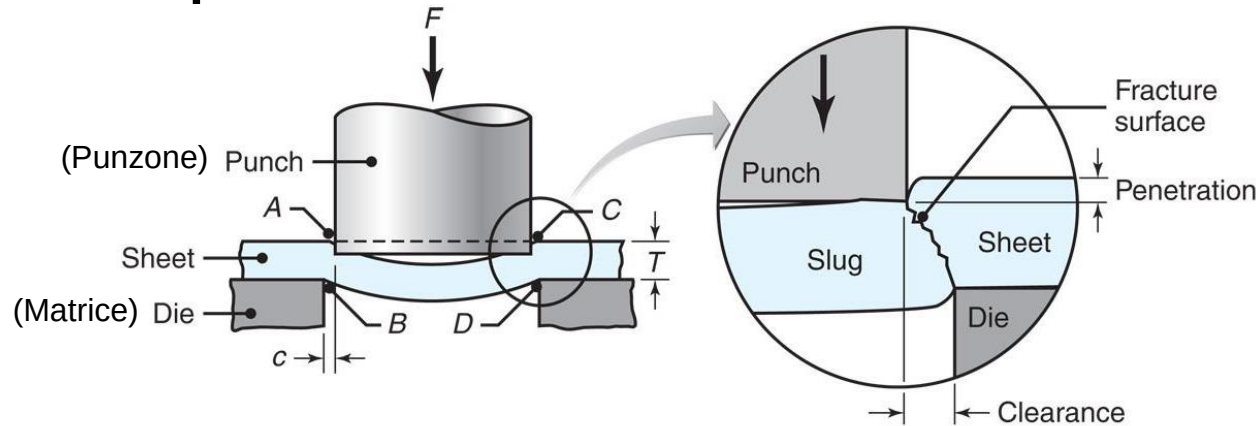
# Lavorazione delle lamiere

## Tranciatura e punzonatura

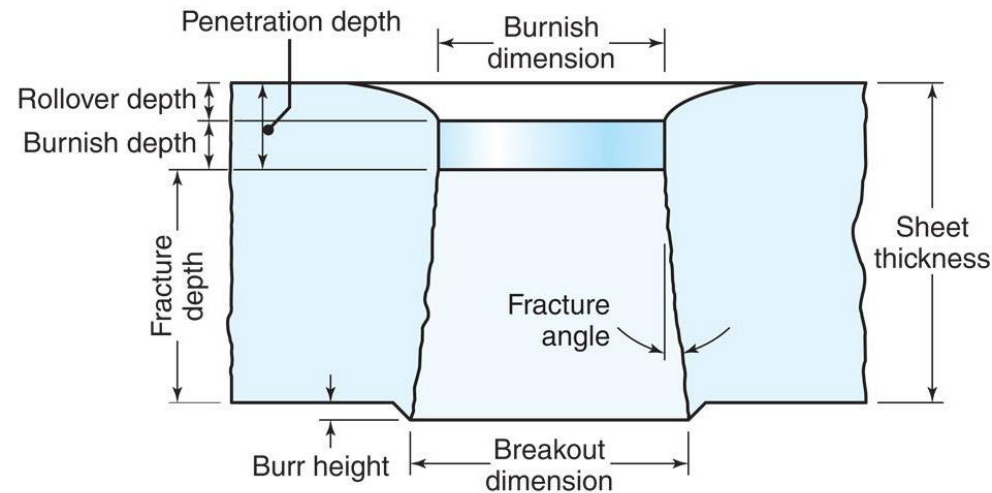


# Lavorazione delle lamiere

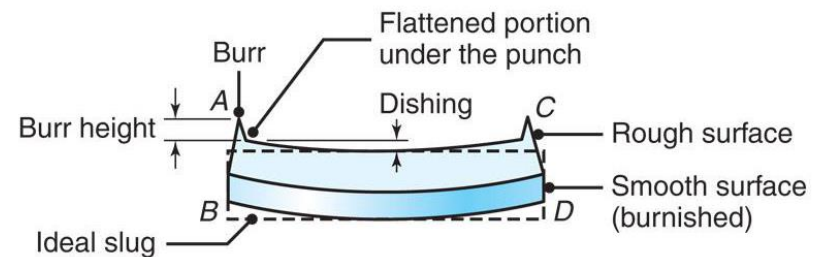
## Tranciatura e punzonatura



Schema del processo

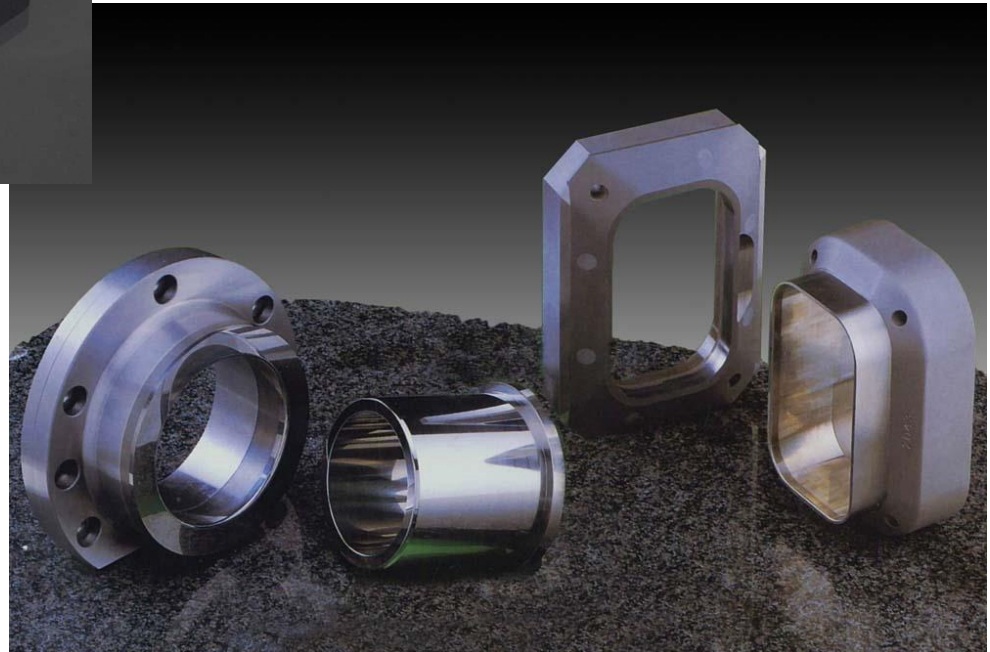
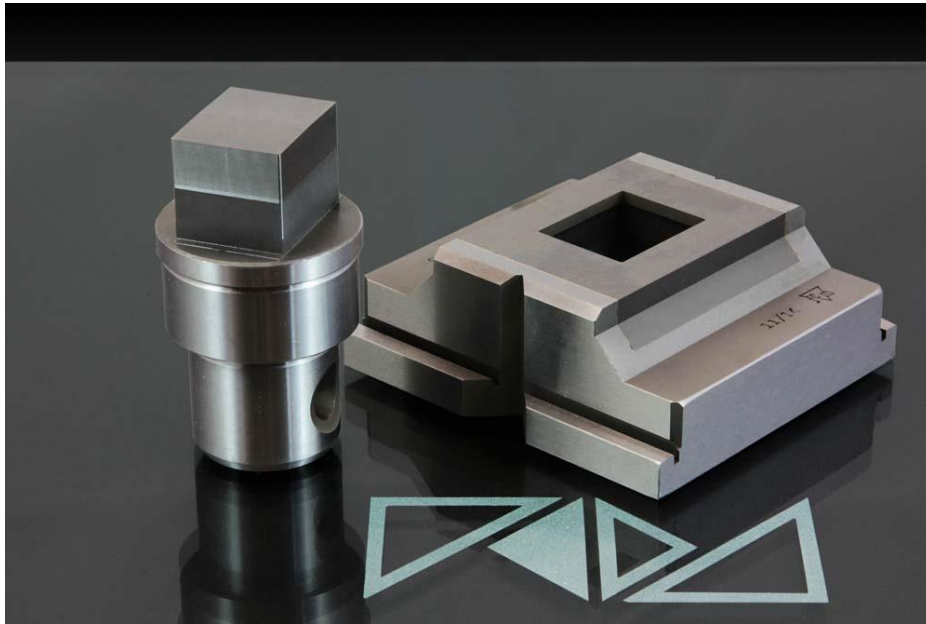


Foro tranciato



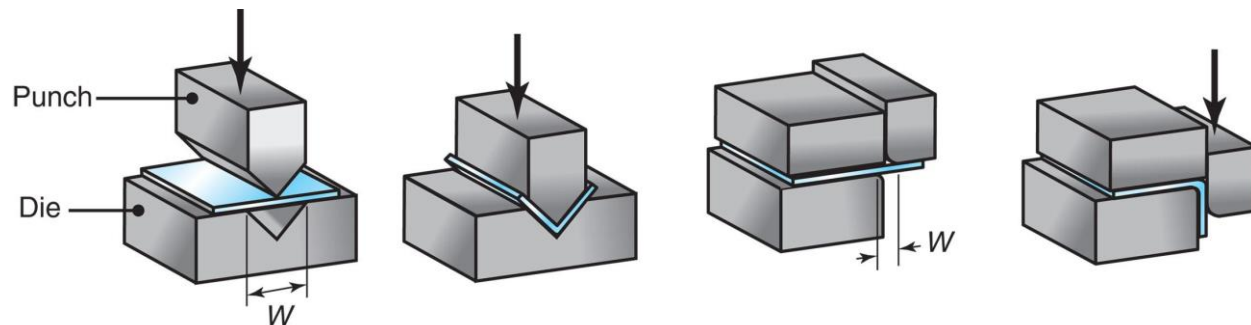
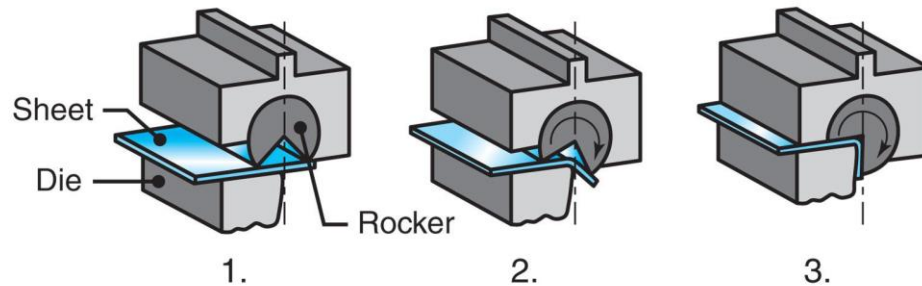
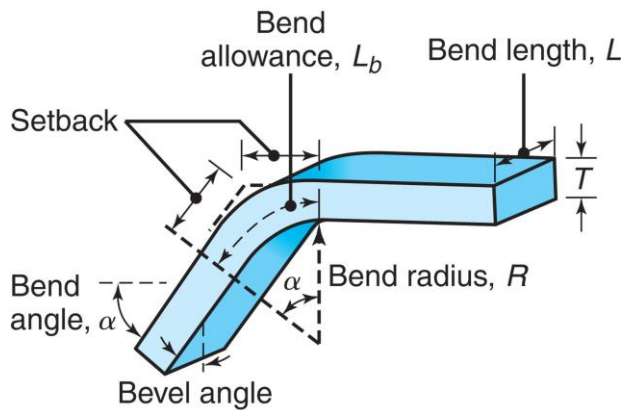
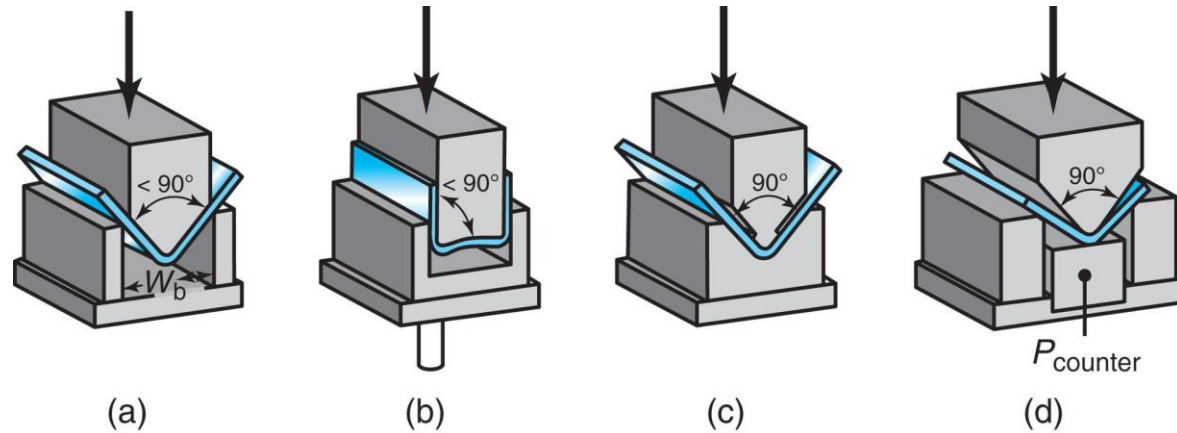
Piattello

# Lavorazione delle lamiere



# Lavorazione delle lamiere

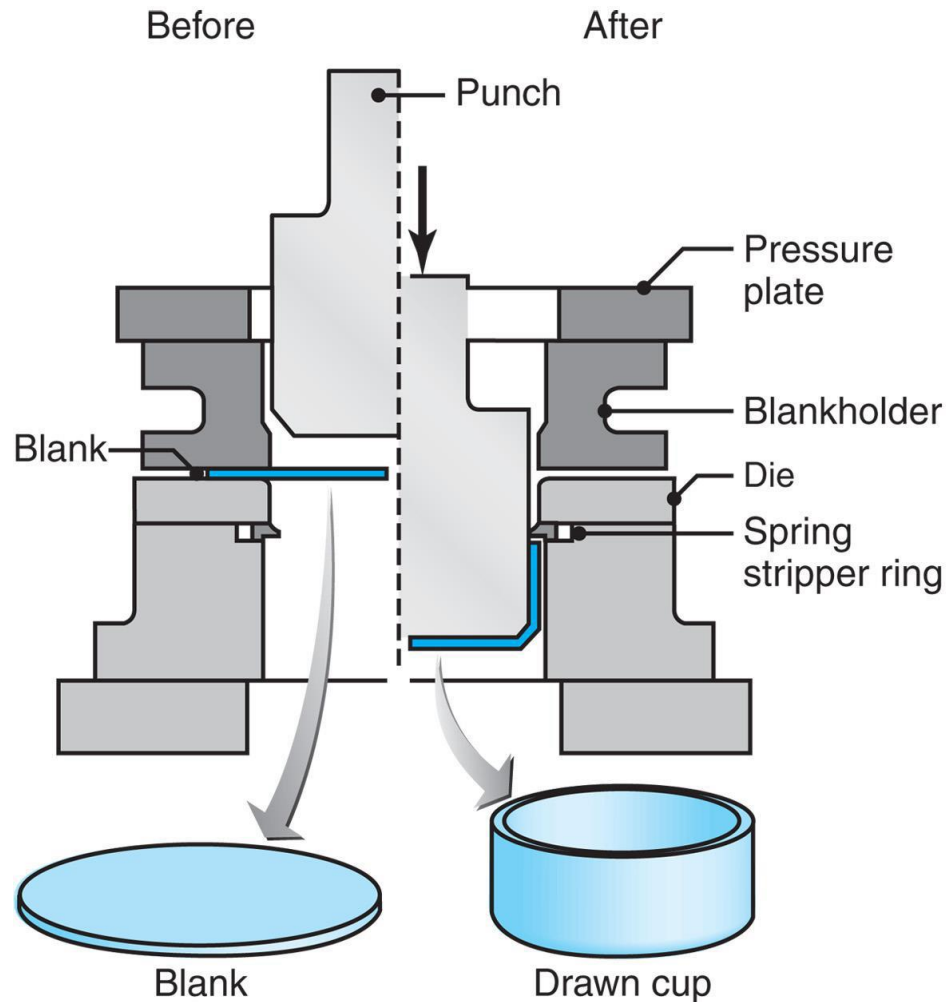
## Piegatura



[VIDEO](#)

# Lavorazione delle lamiere

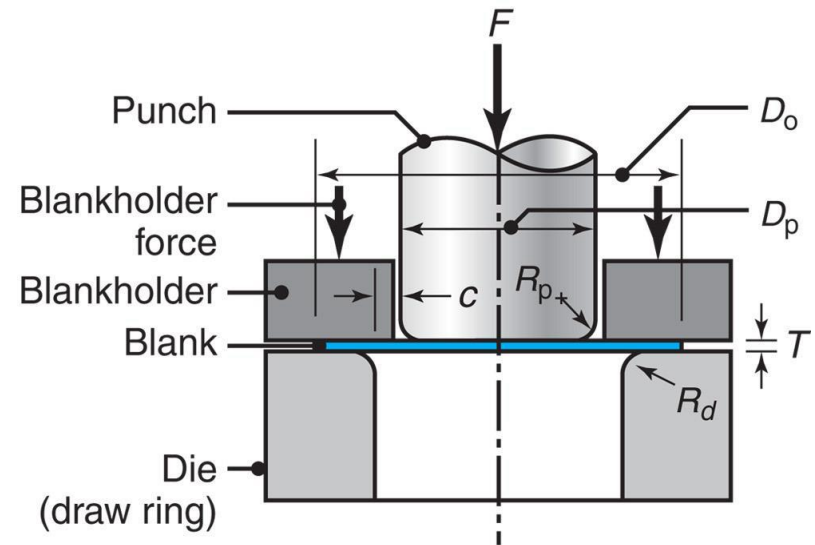
## Imbutitura



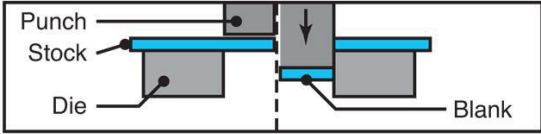
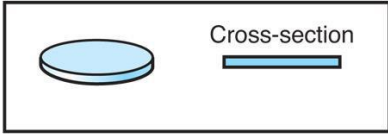
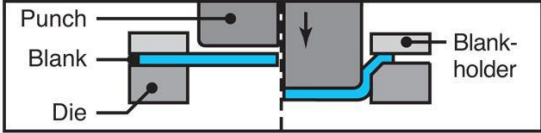
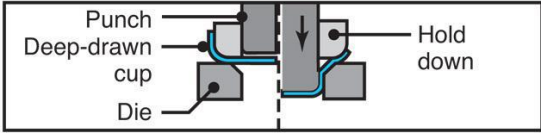
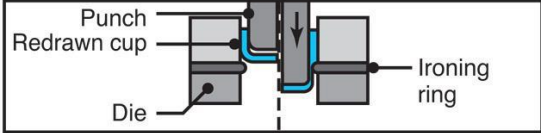
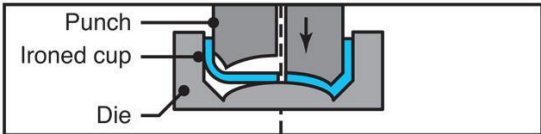
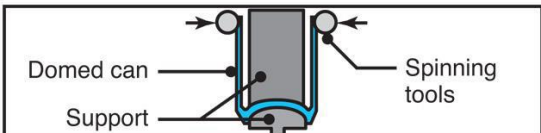
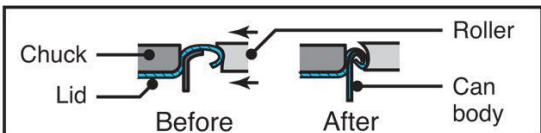
[Video](#)

[Video: lavabo](#)

[Video: portiera auto](#)



# Lavorazione delle lamiere

| Process         | Process illustration                                                                 | Result                                                                                |
|-----------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1. Blanking     |    |    |
| 2. Deep drawing |    |    |
| 3. Redrawing    |    |    |
| 4. Ironing      |    |    |
| 5. Doming       |   |   |
| 6. Necking      |  |  |
| 7. Seaming      |  |  |

# Lavorazione delle lamiere

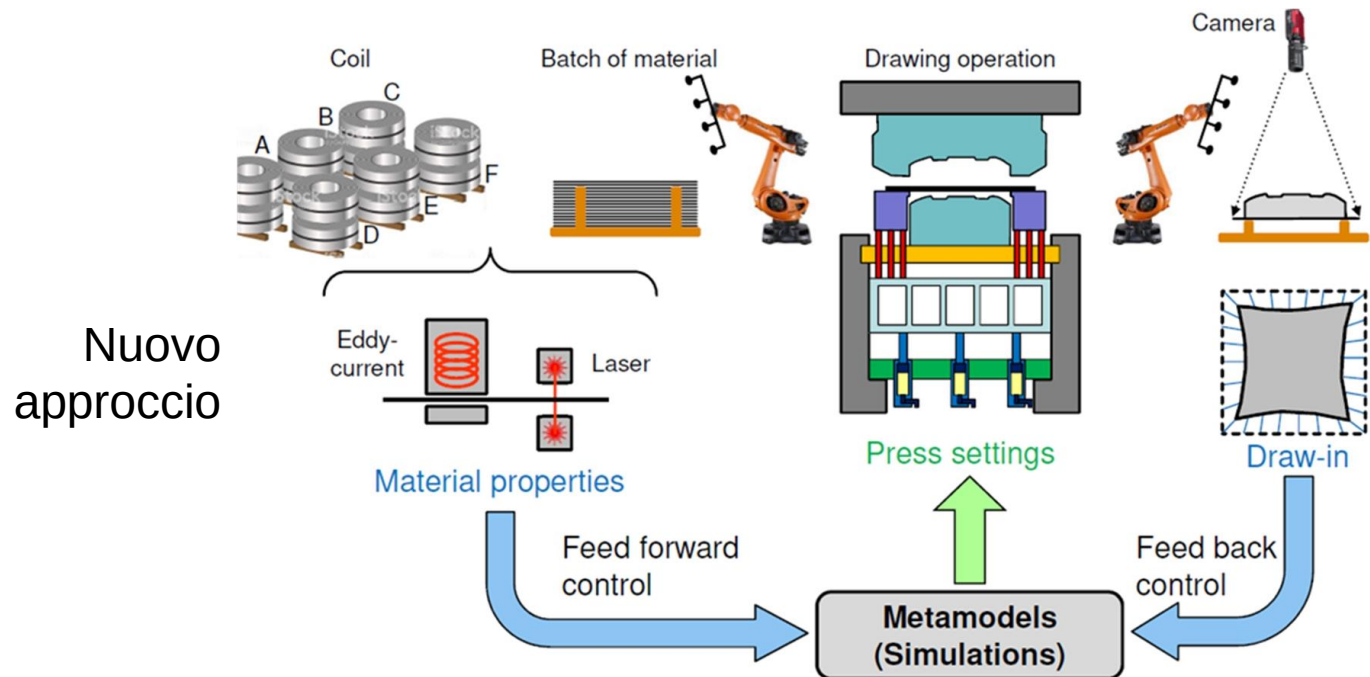
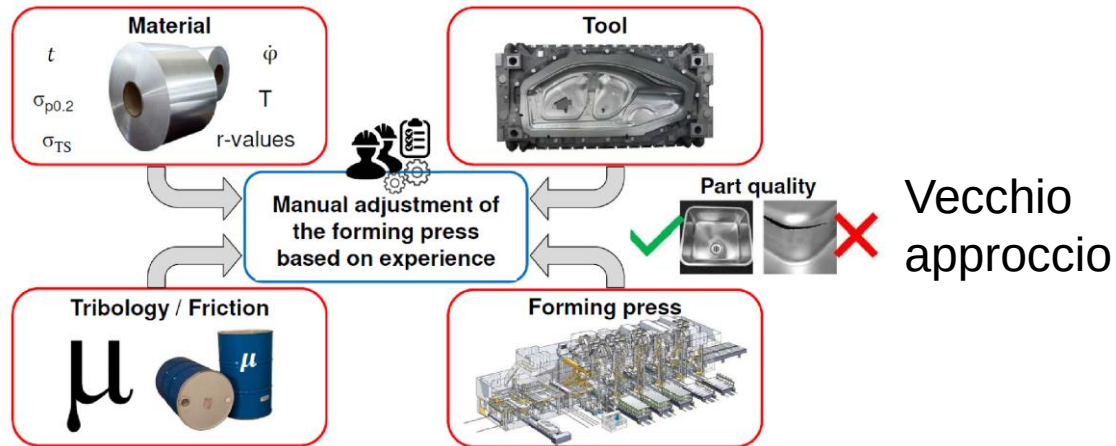


Stampo  
progressivo a  
due impronte

Pressa: 400t

[Source](#)

# Industria 4.0 applicata alle lamiere



[Source](#)



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UNIVERSITÀ DEGLI STUDI DI  
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria  
"Enzo Ferrari"

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